EVERY TWO-COLORING OF THE PLANE CONTAINS A MONOCHROMATIC DEGENERATE TRIANGLE WITH SIDES 1, 3, AND 4

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Abstract. We prove that every two-coloring of the Euclidean plane contains three collinear points of one color with consecutive distances 1 and 3. Suppose, for a contradiction, that some two-coloring contains no such triple. We consider the triangular lattice $\Lambda = \mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$, together with its two rotated copies $q\Lambda$ and $\bar{q}\Lambda$, where $q = \frac{8+5\omega}{7}$. Each copy shares a sublattice with Λ, so the colorings of the three lattices constrain each other. We show that, in one of its three directions, every row of a unit triangular lattice is colored in monochromatic runs of length exactly four, and that consecutive such rows are related by a fixed shift. Then any two points of the plane at distance 8 have the same color, which is impossible under our assumption.

1. Introduction

Let $\mathbb{E}^2$ denote the Euclidean plane. Suppose that we have a coloring of $\mathbb{E}^2$ using two colors. We are interested in three-point configurations that appear monochromatically in every such coloring. We use the term *triple* to refer to any set of three distinct points.

Definition 1.1. The triple $T(a, b, c)$ is a triple of points forming the vertices of a triangle whose sides have lengths a, b, and c. This includes collinear configurations, which correspond to *degenerate* triangles, with one side equal to the sum of the other two.

The notation $T(a)$ is shorthand for $T(a, a, a)$, the vertices of an equilateral triangle of side length a.

Definition 1.2. A coloring *contains* $T(a, b, c)$ exactly when some congruent copy of it is monochromatic.

A foundational contribution to Euclidean Ramsey theory—which studies such questions, as well as more general ones—was made in the 1970s by Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus. In particular, in their 1973 paper [EGM+73], they showed that $T(a)$ can be avoided: it suffices to divide the plane into parallel stripes of width $\frac{a\sqrt{3}}{2}$ and color them alternately. Later, Jelínek, Kynčl, Stolař, and Valla [JKS+09] also found non-trivial colorings avoiding $T(a)$. To this day, these are the only triples for which such a coloring is known.

In 1975, Erdős et al. [EGM+75] identified several families of triples that are contained in every two-coloring of the Euclidean plane. These include configurations $T(a, b, c)$ such that:

- the ratio $\frac{a}{b}$ equals $2\sin\left(\frac{\theta}{2}\right)$ for some $\theta \in \{30°, 36°, 108°, 120°, 150°\}$[1];
- the triangle has an angle of 30° or 150°;
- the triangle has angles $(\alpha, 2\alpha, 180° - 3\alpha)$ for $0 < \alpha < 60°$, $\alpha \neq 45°$;
- the triangle has angles $(180° - \alpha, 180° - 2\alpha, 3\alpha - 180°)$ for $60° < \alpha < 90°$;
- $T(a, b, c) = T(a, 2a, 3a)$, a degenerate triangle;
- a, b, and c satisfy $c^2 = a^2 + 2b^2$, with $b < 2a$;
- a, b, and c satisfy $a^6 - 2a^4b^2 + a^2b^4 - 3a^2b^2c^2 + b^2c^4 = 0$;
- a, b, and c satisfy $a^4c^2 + b^4a^2 + c^4b^2 - 5a^2b^2c^2 = 0$;
- a, b, and c are the base, the leg, and the circumradius of an isosceles triangle.

In the following year, Shader [Sha76] added the triples $T(a, b, c)$ for which

- the triangle is right-angled;
- a, b, and c satisfy $a^2 + c^2 = 4b^2$, where $\frac{\sqrt{7}-1}{2}\, b \leq a \leq \frac{\sqrt{7}+1}{2}\, b$ (Shader's original range for a was narrower and was extended in the author's thesis [Yan25]).

[1] Erdős et al. erroneously mention the angle 72° instead of 36° in their paper.

After these classical results, no new triples of this kind were found for over four decades, until Currier, Moore, and Yip [CMY24] proved that every two-coloring contains $T(1,1,2)$. Their result was an inspiration for this research, in which we prove the analogous statement for $T(1,3,4)$. By scaling, it then holds for every $T(a,3a,4a)$ as well.

Theorem 1.3. *For every $a>0$, every two-coloring of $\mathbb{R}^2$ contains $T(a,3a,4a)$.*

Theorem 1.3 is proved in Section 3. Several lemmas about runs are taken from the author's thesis [Yan25] (Section 2.2), as is the idea of overlapping a lattice with a rotated copy, proposed in its conclusion.

2. Preliminaries

2.1. **Notation and key triples.** The *Eisenstein integers* are the complex numbers $a+b\omega$ with $a,b\in\mathbb{Z}$, where $\omega=e^{\frac{2\pi i}{3}}=\frac{-1+i\sqrt{3}}{2}$ is a primitive cube root of unity. We write $\Lambda=\mathbb{Z}[\omega]=\mathbb{Z}+\mathbb{Z}\omega$ for the set of all of them. Then $\omega^2=-1-\omega$, $\bar{\omega}=\omega^2$, and $|a+b\omega|^2=a^2-ab+b^2$. Viewed as a subset of the plane, Λ is the triangular lattice (Figure 1). For $x\in\mathbb{C}$ write $x\Lambda=\{x\lambda:\lambda\in\Lambda\}$. A *unit triangular lattice* is a set $z_0+u\Lambda$ with $|u|=1$. Its three *directions* are $u,u\omega,u\omega^2$, and its *rows* in a direction are the lines parallel to it.

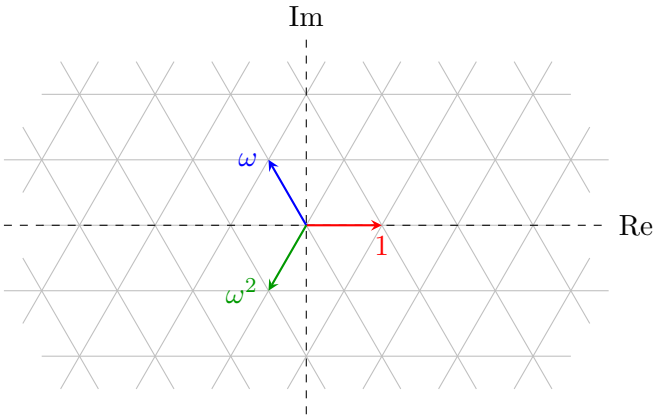


Figure 1. The lattice Λ is spanned by 1 and ω

A two-coloring of the plane is *admissible* if it does not contain $T(1,3,4)$. From now on, we assume that an admissible coloring is given and derive a contradiction. We call the two colors orange and yellow, and write $\chi(p)$ for the color of a point p.

Theorem 2.1 ([EGM+75])**.** *A two-coloring of the plane contains $T(a,b,c)$ if and only if it contains $T(a)$, $T(b)$, or $T(c)$.*

The proof by Erdős et al. [EGM+75] applies equally to degenerate triangles. In particular, for $T(1,3,4)$ it means that a coloring contains $T(1,3,4)$ if and only if it contains $T(1)$, $T(3)$, or $T(4)$.

Definition 2.2. A *key triple* is a congruent copy of $T(1)$, $T(3)$, $T(4)$, or $T(1,3,4)$. In an admissible coloring, no key triple is monochromatic.

2.2. **Runs in a lattice.** We need three facts about a single lattice, all taken from the analysis of $T(1,3,4)$-free colorings in the author's thesis [Yan25, Section 3]. Complete proofs are included here for admissible colorings of the plane. Fix an admissible coloring and a unit triangular lattice L.

Definition 2.3. A *run* is a maximal set of consecutive points of one color on a row.

Lemma 2.4 (runs)**.**

(a) *Every run has at most four points.*
(b) *If a run has at least three points, then the runs next to it on both sides have the other color and at least three points.*

(c) *On each row either every run has at most two points (such a row is called* short*) or every run has three or four points (such a row is called* long*).*

Proof. Identify L with Λ so that the row is $\mathbb{Z}$.

(a) If $0, 1, 2, 3, 4$ have one color, then $\{0, 1, 4\}$ is a monochromatic $T(1, 3, 4)$.

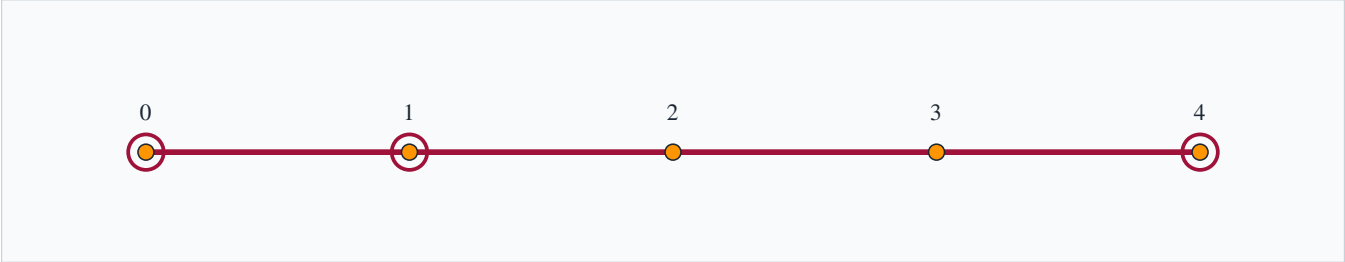


FIGURE 2. Five consecutive points of one color

(b) Let $0, 1, 2$ be the last three points of an orange run, so 3 is yellow. The copy $\{0, 1, 4\}$ of $T(1, 3, 4)$ makes 4 yellow, and the copy $\{1, 2, 5\}$ makes 5 yellow. So the next run is yellow with at least three points. The other side is symmetric.

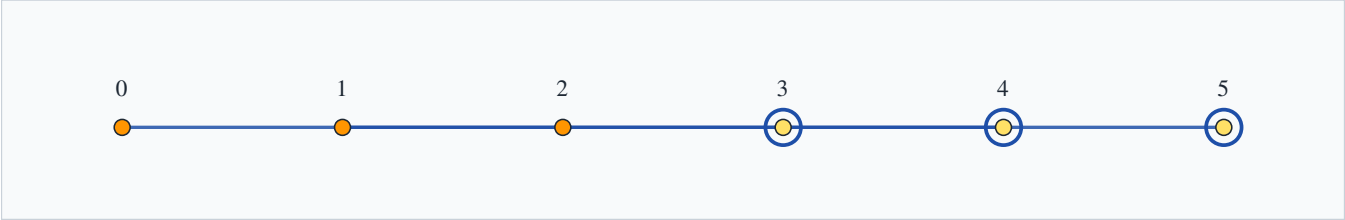


FIGURE 3. The run after an orange run

(c) By (b), if one run on a row has at least three points, so do its neighbors, and by induction all runs on the row. By (a), they have three or four points. □

Lemma 2.5 (adjacent rows)**.** *Two adjacent parallel rows are both long or both short. Hence the rows of one direction are either all long or all short, and we call the direction long or short accordingly.*

Proof. Identify L with Λ so that the rows are $\mathbb{Z}$ and $\mathbb{Z} + \omega$. The reflection $z \mapsto \bar{z} + \omega$ swaps them, so we may assume that $\mathbb{Z}$ is long and $\mathbb{Z} + \omega$ is short. Exchanging colors and translating along the row, let $0, 1, 2$ be the last three points of an orange run of $\mathbb{Z}$. By Lemma 2.4(b), 3 and 4 are yellow. The copy $\{0, 1, 1+\omega\}$ of $T(1)$ makes $1 + \omega$ yellow, and the copy $\{1, 2, 2 + \omega\}$ makes $2 + \omega$ yellow. As $\mathbb{Z} + \omega$ is short, their run is exactly $\{1 + \omega, 2 + \omega\}$, so ω and $3 + \omega$ are orange. The copy $\{3, 4, 4 + \omega\}$ of $T(1)$ makes $4 + \omega$ orange, and ω, $3 + \omega$, $4 + \omega$ form an orange $T(1, 3, 4)$, a contradiction (Figure 4). The second statement follows by induction over the rows of a direction.

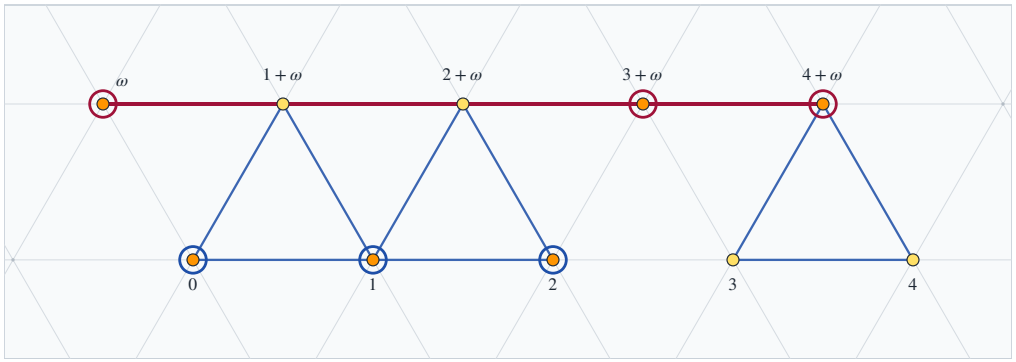


FIGURE 4. A long row below a short row

□

Lemma 2.6 (a long direction)**.** *At least one direction is long.*

Proof. Suppose every row is short, so no three consecutive points of a row have one color. By the pigeonhole principle, two vertices of any unit triangle have the same color. They are adjacent on a row, and after a symmetry of L and exchanging colors we may assume that they are the orange points $0, 1$ of Λ. As rows are short, -1 and 2 are yellow. The copy $\{0, 1, 1+\omega\}$ of $T(1)$ makes $1+\omega$ yellow, and the copy $\{-3, 0, 1\}$ of $T(1, 3, 4)$ makes -3 yellow. Then -2 is orange, since $-3, -2, -1$ lie on a row. Similarly, the copy $\{-1, 2, 2+3\omega\}$ of $T(3)$ makes $2+3\omega$ orange, and the copy $\{-1, 2, 3\}$ of $T(1, 3, 4)$ makes 3 orange. In turn, the copy $\{-2, 1, 1+3\omega\}$ of $T(3)$ forces $1+3\omega$ to be yellow, and the copy $\{0, 3, 3+3\omega\}$ forces $3+3\omega$ to be yellow. The points $1+\omega, 1+2\omega, 1+3\omega$ and $1+\omega, 2+2\omega, 3+3\omega$ lie on rows, so $1+2\omega$ and $2+2\omega$ are orange, and $1+2\omega, 2+2\omega, 2+3\omega$ is an orange $T(1)$, a contradiction (Figure 5).

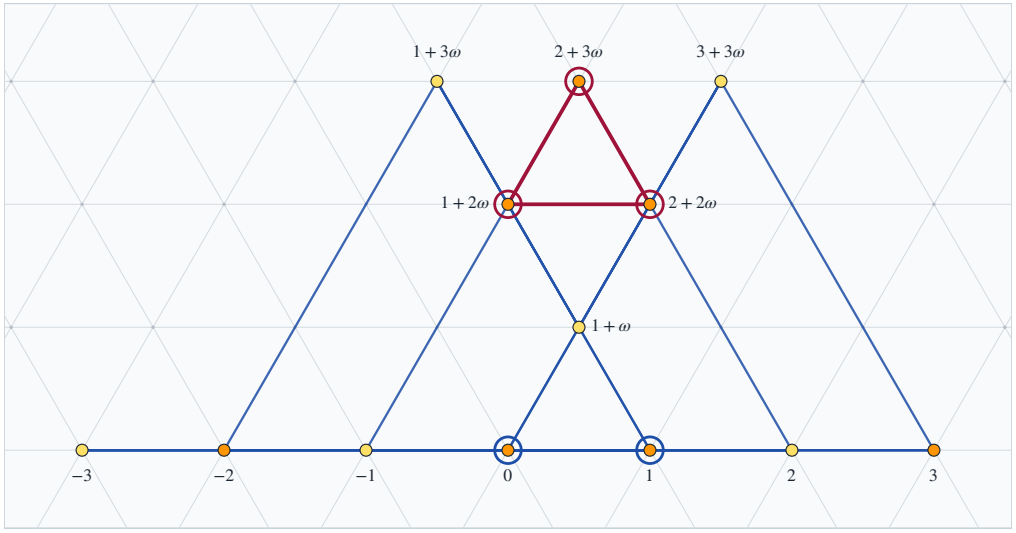


FIGURE 5. If every row is short

□

2.3. **Rotated copies of the lattice.** Section 2.2 alone does not give a contradiction, but it takes away much of the freedom. In an admissible coloring of the plane, every unit triangular lattice has a long direction, and different lattices share points. The idea is therefore to overlap such lattices with many common points (in fact, whole sublattices), so that they constrain each other there.

We want a second unit triangular lattice $q\Lambda$, $|q| = 1$, that is genuinely rotated and shares many points with Λ, since shared points are where constraints pass from one lattice to the other. The points $2-\omega$ and $3+\omega$ of Λ are symmetric about the real axis, so they lie at the same distance from 0. This distance is $\sqrt{7}$,

since $3+\omega = \frac{5}{2} + \frac{\sqrt{3}}{2}\,i$ and $\frac{25}{4} + \frac{3}{4} = 7$. Let q be the rotation about 0 that maps $2-\omega$ to $3+\omega$ (Figure 6). It is the multiplication by the quotient of the two points:

$$q = \frac{3+\omega}{2-\omega} = \frac{8+5\omega}{7} = \frac{11+5\sqrt{3}\,i}{14}, \qquad |q| = 1.$$

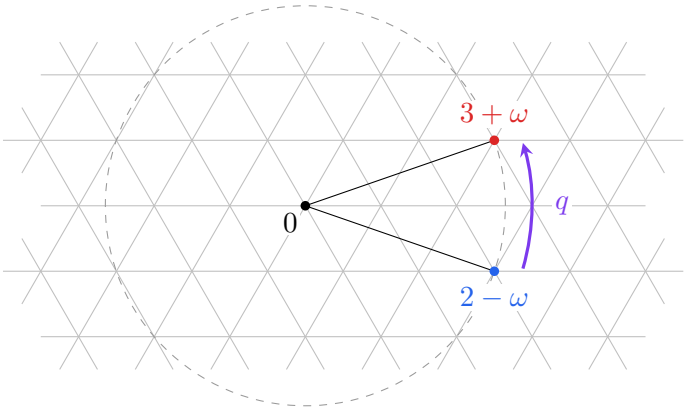


FIGURE 6. The rotation q maps $2-\omega$ to $3+\omega$

Definition 2.7. The rotated lattice is $q\Lambda = \{q(m+n\omega) : m, n \in \mathbb{Z}\}$.

The point $q = q \cdot 1$ lies in $q\Lambda$ but not in Λ, so $q\Lambda \neq \Lambda$. In particular, q is not a sixth root of unity, since all of them lie in Λ. Rotating the multiples of $2-\omega$ gives the multiples of $3+\omega$, as $q\,(2-\omega)\lambda = (3+\omega)\lambda$. Hence the whole sublattice $(3+\omega)\Lambda$ lies in both Λ and $q\Lambda$ (Figure 7). There are no other common points.

Lemma 2.8. $\Lambda \cap q\Lambda = (3+\omega)\Lambda$.

Proof. It remains to show that every point of $\Lambda \cap q\Lambda$ lies in $(3+\omega)\Lambda$. Let $z = q(a+b\omega)$ lie in Λ. As $q = \frac{8+5\omega}{7}$ and $\omega^2 = -1-\omega$,

$$z = \frac{(8a-5b) + (5a+3b)\omega}{7},$$

so 7 divides $8a-5b = 8(a+2b)-21b$, and hence $a+2b$. Let $d = \frac{a+2b}{7}$ and $c = 3d-b$. Then $(2-\omega)(c+d\omega) = (2c+d) + (3d-c)\omega = a+b\omega$, so $z = q(2-\omega)(c+d\omega) = (3+\omega)(c+d\omega)$. □

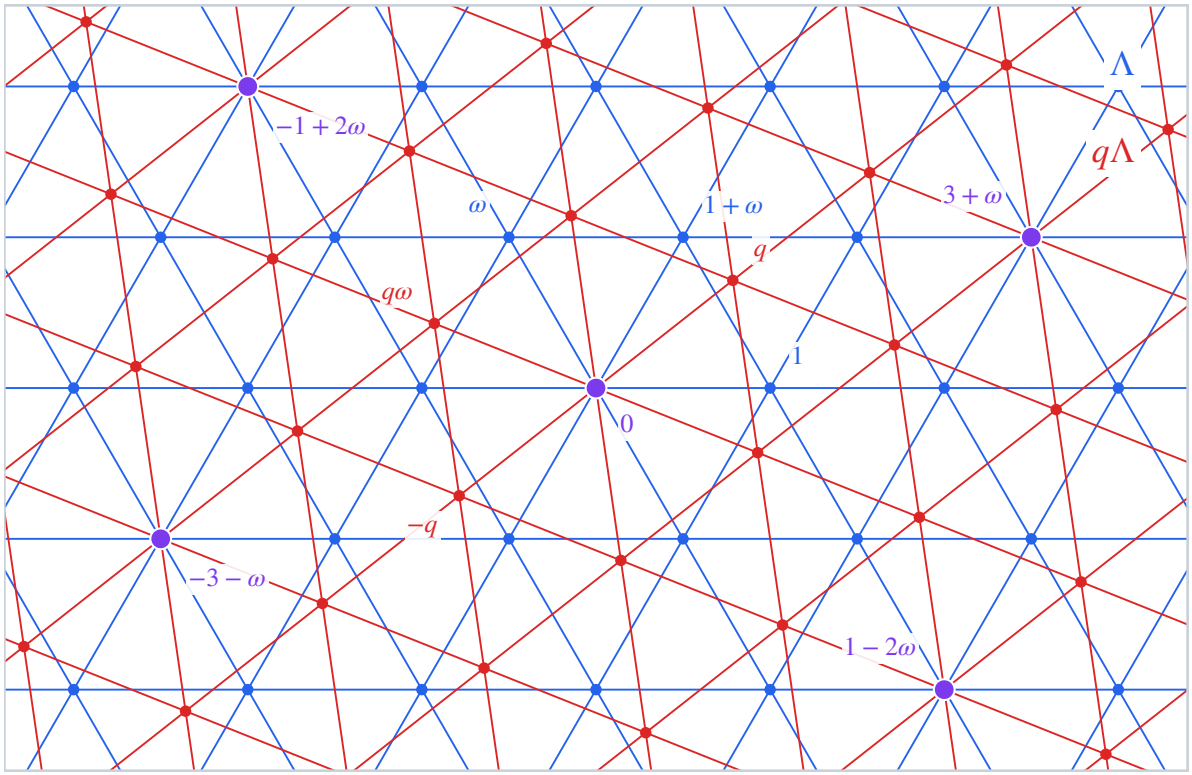


FIGURE 7. Λ (blue), $q\Lambda$ (red), and their common points (purple)

The reflection $z \mapsto \bar{z}$ maps Λ to itself and $q\Lambda$ to $\bar{q}\Lambda$, where $\bar{q} = \frac{8+5\bar{\omega}}{7} = \frac{3-5\omega}{7} = q^{-1}$, so $\bar{q}\Lambda$ is Λ rotated by the opposite angle. By Lemma 2.8, it shares exactly the sublattice $\overline{(3+\omega)\Lambda} = (2-\omega)\Lambda$ with Λ. Its points are written $\bar{q}(m+n\omega)$.

Lemma 2.9. *$q\Lambda \cap \bar{q}\Lambda = 7\Lambda$.*

Proof. As $7 = q(3-5\omega) = \bar{q}(8+5\omega)$, the sublattice 7Λ lies in both $q\Lambda$ and $\bar{q}\Lambda$. Conversely, let $z = q(a+b\omega) = \bar{q}(c+d\omega)$. Expanding both products gives

$$7z = (8a-5b) + (5a+3b)\omega = (3c+5d) + (8d-5c)\omega.$$

Let $X = a+2b$ and $Y = c+4d$. Modulo 7, the two coordinates are $X \equiv 3Y$ and $5X \equiv 2Y$. Hence $2Y \equiv 5X \equiv 15Y \equiv Y$, so $Y \equiv 0$ and $X \equiv 0$. Both coordinates of $7z$ are then divisible by 7, so $z \in \Lambda$.

By Lemma 2.8 and its mirror image, $z = m+n\omega$ lies in $(3+\omega)\Lambda$ and in $(2-\omega)\Lambda$. As $(3+\omega)(e+f\omega) = (3e-f) + (e+2f)\omega$ and $(2-\omega)(e+f\omega) = (2e+f) + (3f-e)\omega$, the numbers $m-3n$ and $m+2n$ are multiples of 7. So 7 divides their difference $5n$, hence n and then m, and $z \in 7\Lambda$. □

Adding to Λ two rotated copies, $q\Lambda$ and its mirror image $\bar{q}\Lambda$, is already enough to derive Theorem 1.3.

3. Proof of Theorem 1.3

3.1. No runs of exactly three.

Theorem 3.1. *In an admissible coloring of the plane, no line of a unit triangular lattice carries a maximal monochromatic run of exactly three points.*

Lemma 3.2 (the row)**.** *If such a run exists then, after an isometry and possibly exchanging colors, $0, 1, 2$ are orange and $-3, -2, -1$ and $3, 4, 5$ are yellow.*

Proof. Normalize the run to $0, 1, 2$ orange with $-1, 3$ yellow, in a lattice Λ. By Lemma 2.4(c) the row $\mathbb{Z}$ is long, so the yellow runs through -1 and 3 have at least three points and contain $-3, -2, -1$ and $3, 4, 5$.

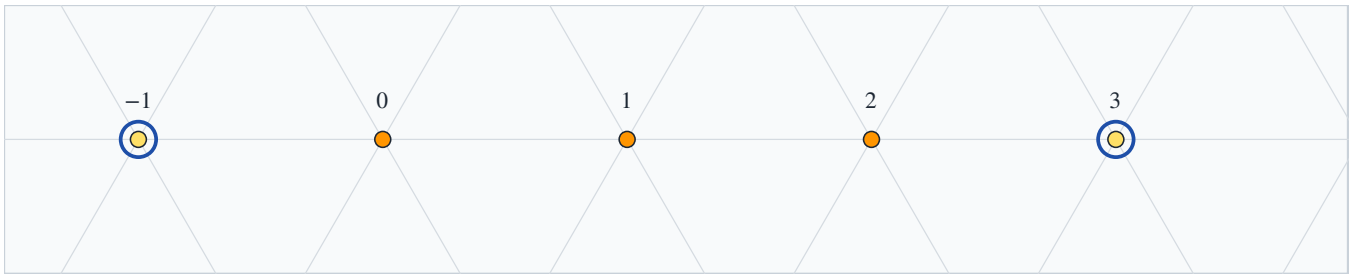


FIGURE 8. The run $0, 1, 2$



FIGURE 9. The neighboring yellow runs

□

Lemma 3.3 (the neighboring rows)**.** *In this situation* $-\omega, 1-\omega, 1+\omega, 2+\omega$ *are yellow, and each of the pairs* $\{-1-\omega, 2-\omega\}$ *and* $\{\omega, 3+\omega\}$ *contains a yellow point.*

Proof. The copies $\{0, 1, -\omega\}$, $\{1, 2, 1-\omega\}$, $\{0, 1, 1+\omega\}$, $\{1, 2, 2+\omega\}$ of $T(1)$ have two orange vertices. By Lemma 2.5 the rows $\mathbb{Z} \pm \omega$ are long, like $\mathbb{Z}$, so the yellow runs through $-\omega, 1-\omega$ and through $1+\omega, 2+\omega$ have length at least three.

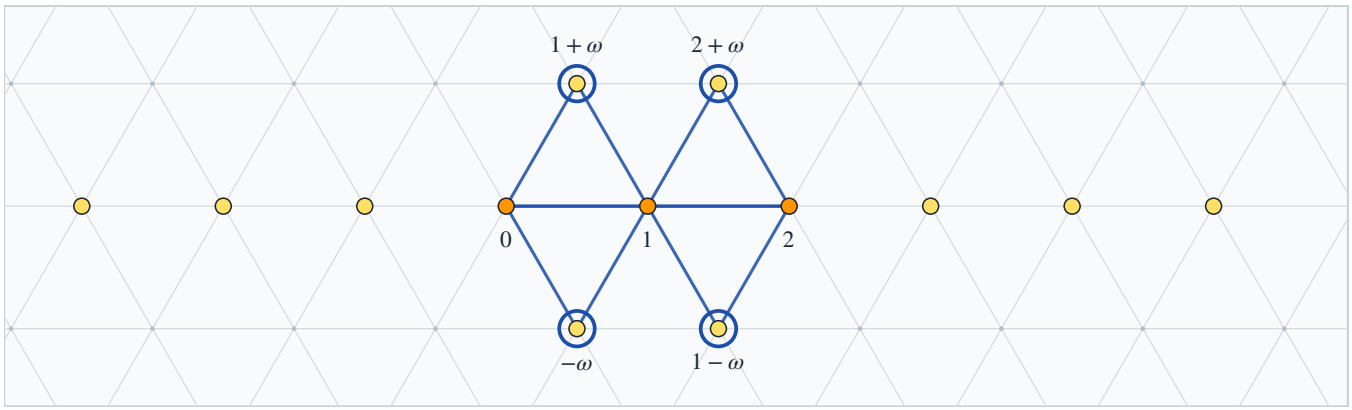


FIGURE 10. Four copies of $T(1)$

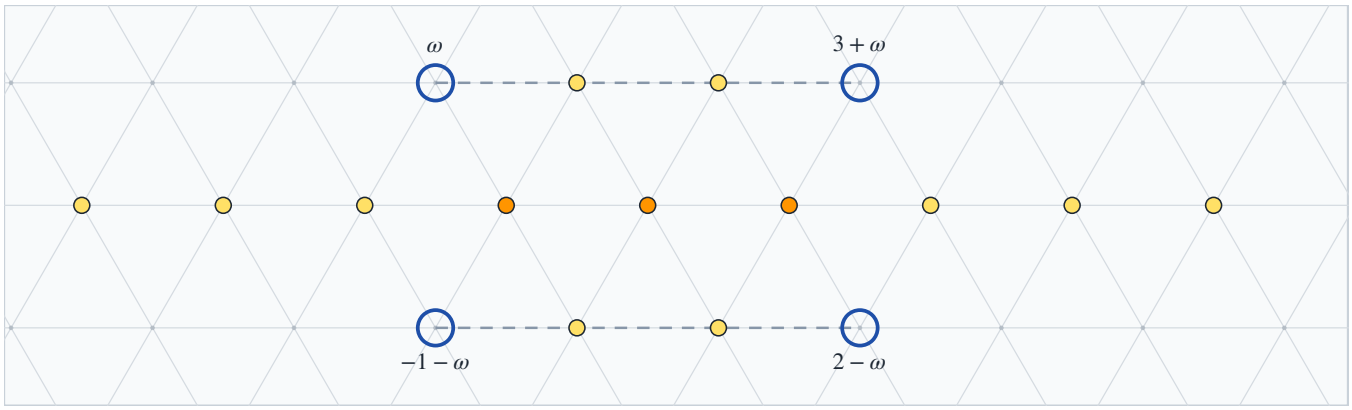


FIGURE 11. The pairs $\{-1-\omega, 2-\omega\}$ and $\{\omega, 3+\omega\}$ (ringed)

□

In particular, 1 is a run of length one on the rows $1-\omega, 1, 1+\omega$ and $-\omega, 1, 2+\omega$. By Lemmas 2.4(c) and 2.5, the directions of ω and $1+\omega$ are therefore short: no three consecutive points in these directions have one color.

Let us call the four points $-1-\omega$, $2-\omega$, ω, and $3+\omega$ (ringed in Figure 11) the *corner points*.

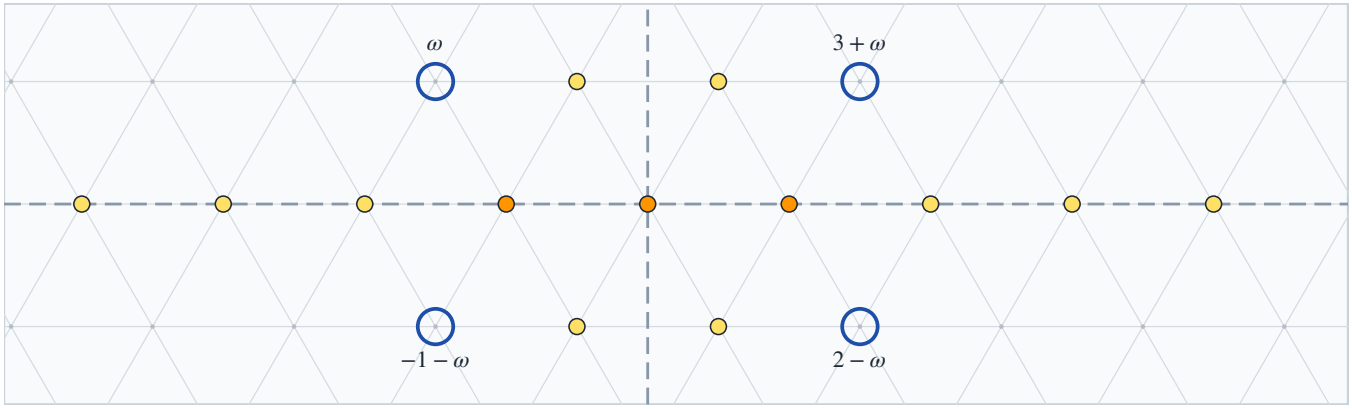


FIGURE 12. The symmetries of the row and the corner points

Proof of Theorem 3.1. By Lemma 3.3, each of the pairs $\{-1-\omega, 2-\omega\}$ and $\{\omega, 3+\omega\}$ contains a yellow point. The row $-3, \dots, 5$ with its colors and the four corner points are symmetric about the real axis and about the vertical line through 1 (Figure 12). A mirror image of an admissible coloring is again admissible, so we may replace the coloring by its mirror image in either line. This swaps the two pairs, or the left and the right corner points. If $2-\omega$ and $3+\omega$ are yellow, we are in Case A1. If $-1-\omega$ and ω are yellow, the reflection in the vertical line leads to Case A1. Otherwise one left and one right corner point are orange. They are not in the same pair, so they are diagonally opposite, and the other two are yellow (Case A2). We now refute both cases inside Λ, $q\Lambda$, and $\bar{q}\Lambda$. In each case, two claims are proved by contradiction, each for a point and, by a symmetry, for its image, and then a short argument ends the case. Every step is shown in a picture. Steps within a claim or at the end of a case use the assumptions of the case and the claims before them.

Markers. By Lemma 2.9, no point lies in $q\Lambda$ and $\bar{q}\Lambda$ without lying in Λ. Hence every point lies in one lattice, in Λ and one of $q\Lambda$, $\bar{q}\Lambda$, or in all three. In the figures, points of Λ are circles, points of $q\Lambda$ upward triangles, and points of $\bar{q}\Lambda$, the mirror image of $q\Lambda$, downward triangles. A point of Λ that also lies in $q\Lambda$ or in $\bar{q}\Lambda$ is a circle with the upward or downward triangle inscribed. The points of all three lattices, that is, the points of 7Λ, are eight-pointed stars (a square and a diamond overlaid). All filled markers have the same area.

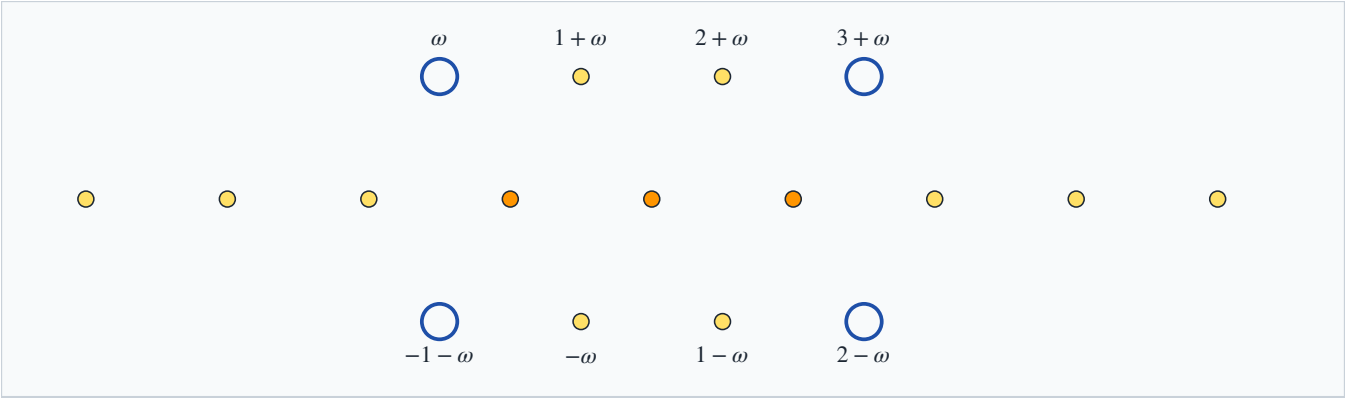


FIGURE 13. The setup of Theorem 3.1

Names. Points of Λ are written $a+b\omega$, points of $q\Lambda$ as $q(m+n\omega)$, and points of $\bar{q}\Lambda$ as $\bar{q}(m+n\omega)$, without parentheses when $m+n\omega$ is a single term, as in $-q\omega$ or $3q$, with $q=\frac{8+5\omega}{7}$ and $\bar{q}=\frac{3-5\omega}{7}$ (Section 2.3).

Steps. Every step is forced by a key triple: two of its points have the same color, so the third point has the other color. Each step names its triple and the two points of equal color, and a check follows. The check lists the three differences of the points, computed in the lattice that contains the triple. Every difference is k, $k\omega$, or $k(1+\omega)$ for an integer k, possibly multiplied by q or $\bar{q}$, so its length is $|k|$ (Figure 14). A point of Λ in a triple of $q\Lambda$ or $\bar{q}\Lambda$ is first renamed in the form of that lattice, as in $7=q(3-5\omega)$. The renaming is checked by multiplying out with $\omega^2=-1-\omega$, as in $(8+5\omega)(3-5\omega)=49$. This form only serves to simplify the check.

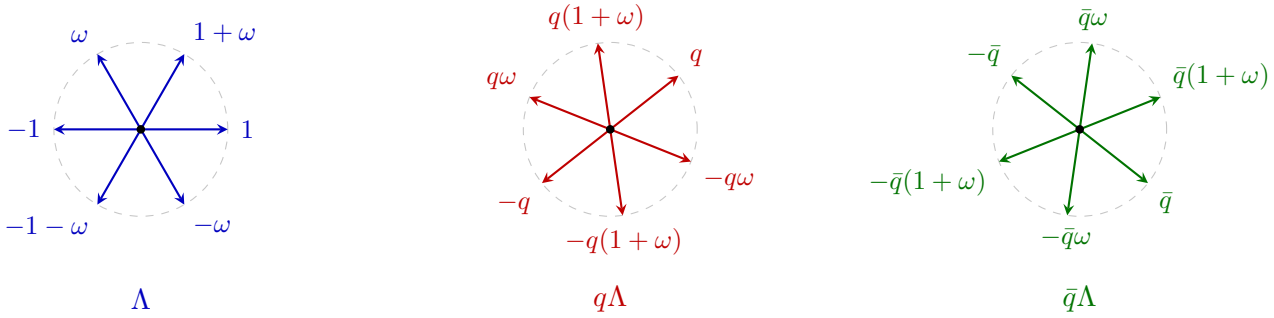


FIGURE 14. The six unit vectors of each of Λ, $q\Lambda$, and $\bar{q}\Lambda$

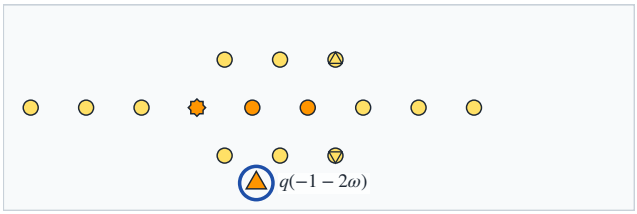
Case A1: assume $2-\omega$ and $3+\omega$ are yellow.

The reflection $z\mapsto\bar{z}$ in the real axis maps the row with its colors, the points $-\omega, 1-\omega, 1+\omega, 2+\omega$, and the two corner points $2-\omega$, $3+\omega$ to themselves (it swaps $-\omega$ with $1+\omega$, $1-\omega$ with $2+\omega$, and $2-\omega$ with $3+\omega$), and it maps $q\Lambda$ onto $\bar{q}\Lambda$. So whenever the assumptions made so far are invariant under it, a statement proved from them also holds for its mirror image: apply the same argument to the coloring $z\mapsto\chi(\bar{z})$, where χ is the given coloring. It is admissible, since the reflection is an isometry.

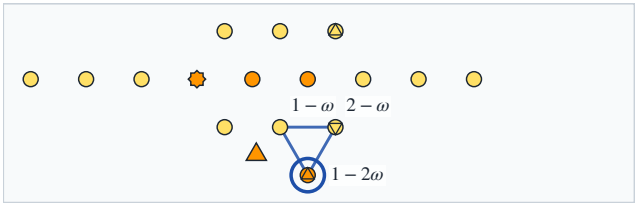


Claim. $q(-1-2\omega)$ and $\bar{q}(1+2\omega)$ are yellow.

Proof.

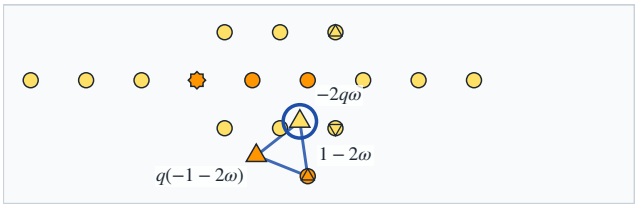


Suppose $q(-1-2\omega)$ is orange.



Step 1. $1-2\omega$ is orange: $T(1)$ with yellow $1-\omega$ and $2-\omega$.
Check:

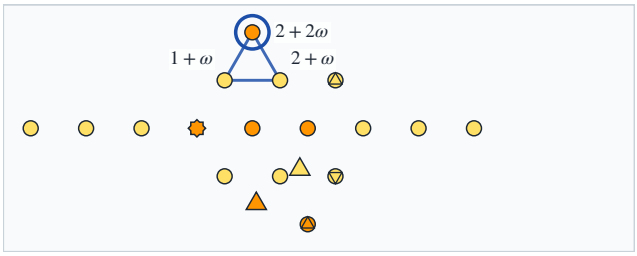
$(1-2\omega)-(1-\omega)=-\omega$
$(1-2\omega)-(2-\omega)=-(1+\omega)$
$(1-\omega)-(2-\omega)=-1$



Step 2. $-2q\omega$ is yellow: $T(1)$ with orange $q(-1-2\omega)$ and $1-2\omega = q(-1-3\omega)$.
Check:

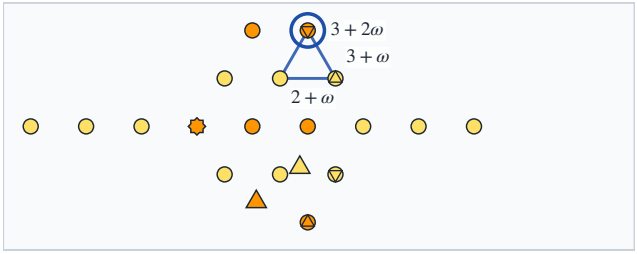
$$q(-1-3\omega)=\frac{(8+5\omega)(-1-3\omega)}{7}=\frac{7-14\omega}{7}=1-2\omega$$

$(-2q\omega)-q(-1-2\omega)=q$
$(-2q\omega)-q(-1-3\omega)=q(1+\omega)$
$q(-1-2\omega)-q(-1-3\omega)=q\omega$



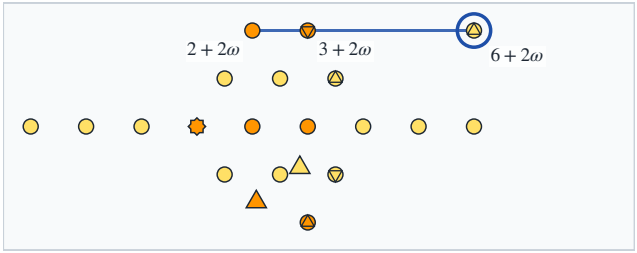
Step 3. $2+2\omega$ is orange: $T(1)$ with yellow $1+\omega$ and $2+\omega$.
Check:

$(2+2\omega)-(1+\omega)=1+\omega$
$(2+2\omega)-(2+\omega)=\omega$
$(1+\omega)-(2+\omega)=-1$



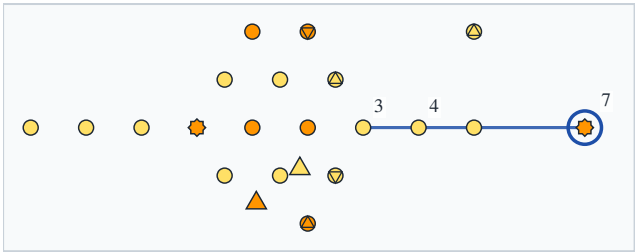
Step 4. $3+2\omega$ is orange: $T(1)$ with yellow $2+\omega$ and $3+\omega$.
Check:

$(3+2\omega)-(2+\omega)=1+\omega$
$(3+2\omega)-(3+\omega)=\omega$
$(2+\omega)-(3+\omega)=-1$

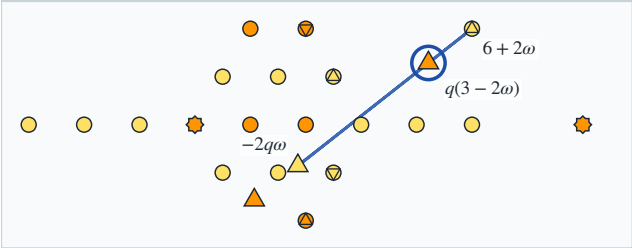


Step 5. $6+2\omega$ is yellow: $T(1,3,4)$ with orange $2+2\omega$ and $3+2\omega$.
Check:

$(6+2\omega)-(2+2\omega)=4$
$(6+2\omega)-(3+2\omega)=3$
$(2+2\omega)-(3+2\omega)=-1$



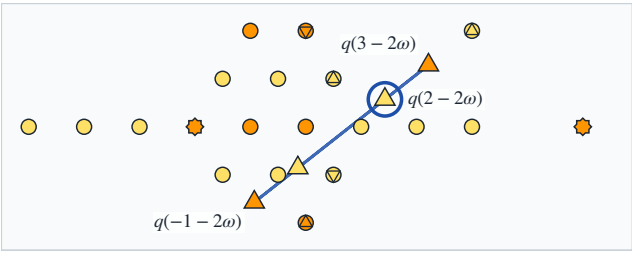
Step 6. 7 is orange: $T(1,3,4)$ with yellow 3 and 4.



Step 7. $q(3-2\omega)$ is orange: $T(1,3,4)$ with yellow $-2q\omega$ and $6+2\omega = q(4-2\omega)$.

Check:

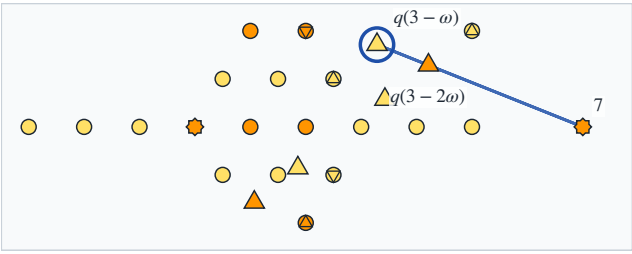
$$q(4-2\omega) = \frac{(8+5\omega)(4-2\omega)}{7} = \frac{42+14\omega}{7} = 6+2\omega$$
$$q(3-2\omega) - (-2q\omega) = 3q$$
$$q(3-2\omega) - q(4-2\omega) = -q$$
$$(-2q\omega) - q(4-2\omega) = -4q$$



Step 8. $q(2-2\omega)$ is yellow: $T(1,3,4)$ with orange $q(-1-2\omega)$ and $q(3-2\omega)$.

Check:

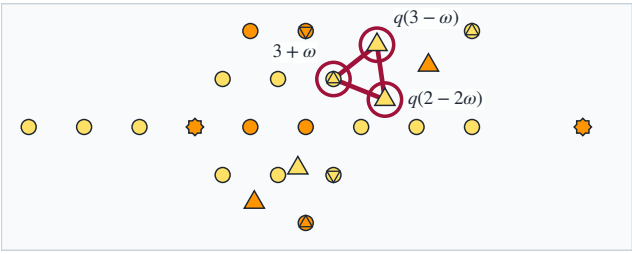
$$q(2-2\omega) - q(-1-2\omega) = 3q$$
$$q(2-2\omega) - q(3-2\omega) = -q$$
$$q(-1-2\omega) - q(3-2\omega) = -4q$$



Step 9. $q(3-\omega)$ is yellow: $T(1,3,4)$ with orange $q(3-2\omega)$ and $7 = q(3-5\omega)$.

Check:

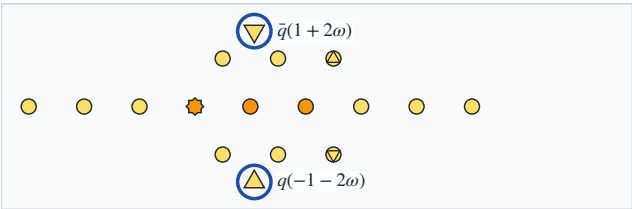
$$q(3-5\omega) = \frac{(8+5\omega)(3-5\omega)}{7} = \frac{49}{7} = 7$$
$$q(3-\omega) - q(3-2\omega) = q\omega$$
$$q(3-\omega) - q(3-5\omega) = 4q\omega$$
$$q(3-2\omega) - q(3-5\omega) = 3q\omega$$



Contradiction: $3+\omega = q(2-\omega)$, $q(2-2\omega)$, $q(3-\omega)$ are all yellow and form $T(1)$.

Check:

$$q(2-\omega) = \frac{(8+5\omega)(2-\omega)}{7} = \frac{21+7\omega}{7} = 3+\omega$$
$$q(2-\omega) - q(2-2\omega) = q\omega$$
$$q(2-\omega) - q(3-\omega) = -q$$
$$q(2-2\omega) - q(3-\omega) = -q(1+\omega)$$



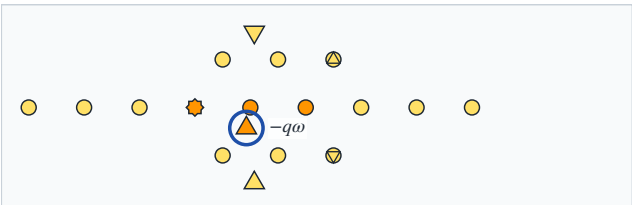
Claim: $q(-1-2\omega)$ and $\bar{q}(1+2\omega)$ are yellow

So $q(-1-2\omega)$ is yellow.

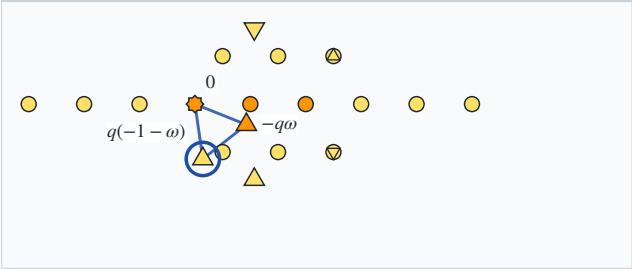
The same argument, applied to the coloring $z \mapsto \chi(\bar{z})$, shows that $\bar{q}(1+2\omega)$ is yellow. □

Claim. $-q\omega$ and $\bar{q}(1+\omega)$ are yellow.

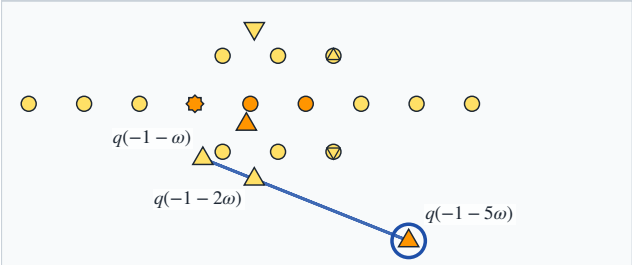
Proof.



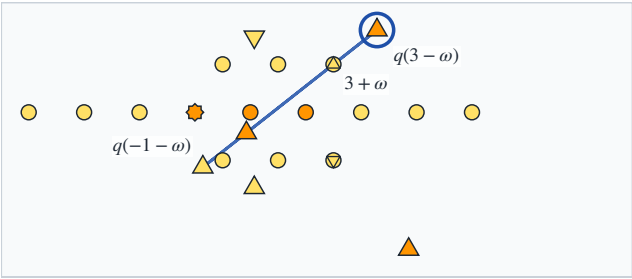
Suppose $-q\omega$ is orange.



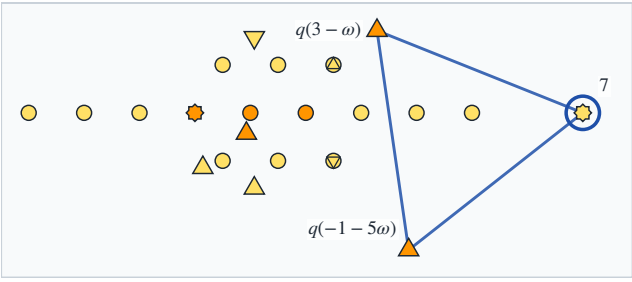
Step 1. $q(-1-\omega)$ is yellow: $T(1)$ with orange 0 and $-q\omega$.
Check:
$q(-1-\omega) - 0 = -q(1+\omega)$
$q(-1-\omega) - (-q\omega) = -q$
$0 - (-q\omega) = q\omega$



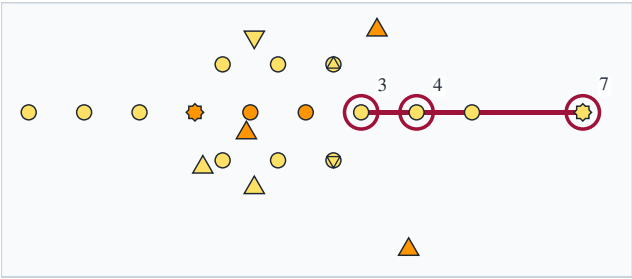
Step 2. $q(-1-5\omega)$ is orange: $T(1,3,4)$ with yellow $q(-1-\omega)$ and $q(-1-2\omega)$.
Check:
$q(-1-5\omega) - q(-1-\omega) = -4q\omega$
$q(-1-5\omega) - q(-1-2\omega) = -3q\omega$
$q(-1-\omega) - q(-1-2\omega) = q\omega$



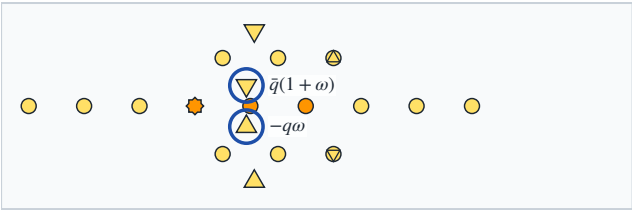
Step 3. $q(3-\omega)$ is orange: $T(1,3,4)$ with yellow $q(-1-\omega)$ and $3+\omega = q(2-\omega)$.
Check:
$q(2-\omega) = \dfrac{(8+5\omega)(2-\omega)}{7} = \dfrac{21+7\omega}{7} = 3+\omega$
$q(3-\omega) - q(-1-\omega) = 4q$
$q(3-\omega) - q(2-\omega) = q$
$q(-1-\omega) - q(2-\omega) = -3q$



Step 4. $7 = q(3-5\omega)$ is yellow: $T(4)$ with orange $q(-1-5\omega)$ and $q(3-\omega)$.
Check:
$q(3-5\omega) = \dfrac{(8+5\omega)(3-5\omega)}{7} = \dfrac{49}{7} = 7$
$q(3-5\omega) - q(-1-5\omega) = 4q$
$q(3-5\omega) - q(3-\omega) = -4q\omega$
$q(-1-5\omega) - q(3-\omega) = -4q(1+\omega)$



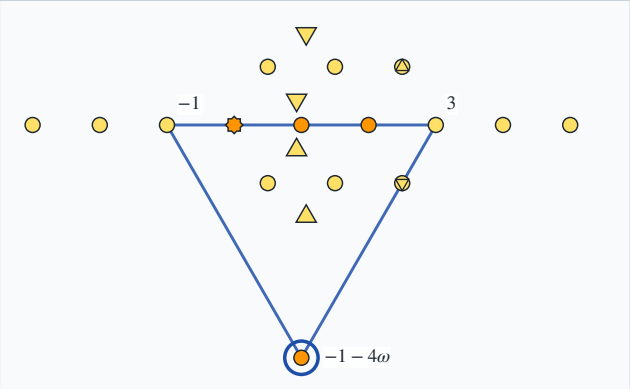
Contradiction: 3, 4, 7 are all yellow and form $T(1,3,4)$.



Claim: $-q\omega$ and $\bar{q}(1+\omega)$ are yellow

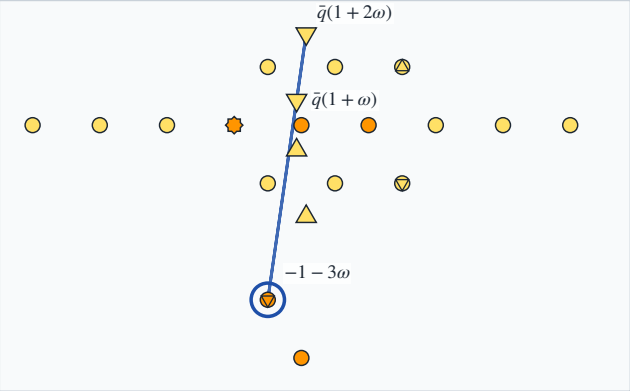
So $-q\omega$ is yellow.
The same argument, applied to the coloring $z \mapsto \chi(\bar{z})$, shows that $\bar{q}(1+\omega)$ is yellow. □

End of Case A1.



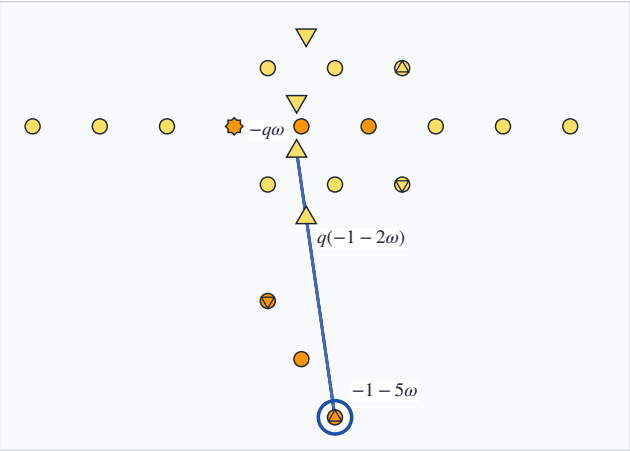
Step 1. $-1-4\omega$ is orange: $T(4)$ with yellow -1 and 3.
Check:

$(-1-4\omega)-(-1)=-4\omega$
$(-1-4\omega)-3=-4(1+\omega)$
$(-1)-3=-4$



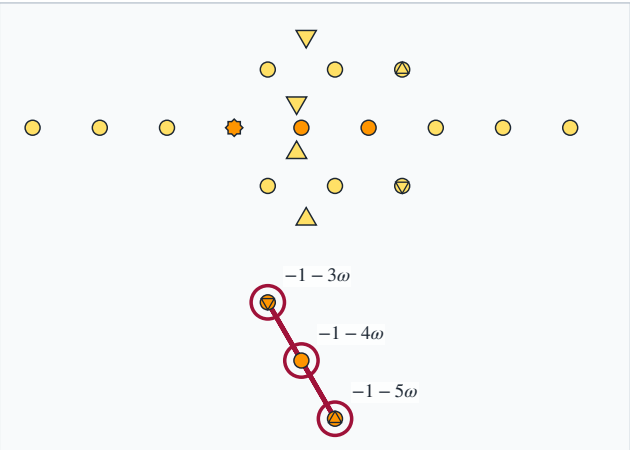
Step 2. $-1-3\omega=\bar{q}(1-2\omega)$ is orange: $T(1,3,4)$ with yellow $\bar{q}(1+\omega)$ and $\bar{q}(1+2\omega)$.
Check:

$\bar{q}(1-2\omega)=\dfrac{(3-5\omega)(1-2\omega)}{7}=\dfrac{-7-21\omega}{7}=-1-3\omega$
$\bar{q}(1-2\omega)-\bar{q}(1+\omega)=-3\bar{q}\omega$
$\bar{q}(1-2\omega)-\bar{q}(1+2\omega)=-4\bar{q}\omega$
$\bar{q}(1+\omega)-\bar{q}(1+2\omega)=-\bar{q}\omega$



Step 3. $-1-5\omega=q(-4-5\omega)$ is orange: $T(1,3,4)$ with yellow $q(-1-2\omega)$ and $-q\omega$.
Check:

$q(-4-5\omega)=\dfrac{(8+5\omega)(-4-5\omega)}{7}=\dfrac{-7-35\omega}{7}=-1-5\omega$
$q(-4-5\omega)-q(-1-2\omega)=-3q(1+\omega)$
$q(-4-5\omega)-(-q\omega)=-4q(1+\omega)$
$q(-1-2\omega)-(-q\omega)=-q(1+\omega)$

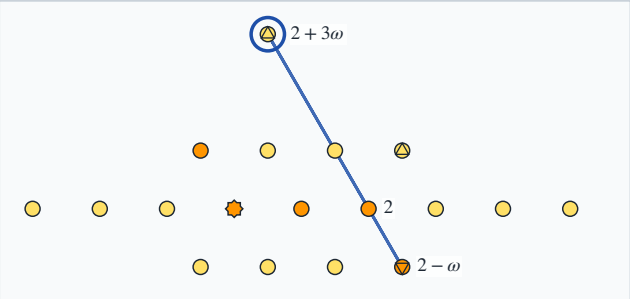


Contradiction: $-1-5\omega$, $-1-4\omega$, $-1-3\omega$ are all orange, three consecutive points of a row in the direction of ω, which is short.
Check:

$(-1-5\omega)-(-1-4\omega)=-\omega$
$(-1-5\omega)-(-1-3\omega)=-2\omega$
$(-1-4\omega)-(-1-3\omega)=-\omega$

Case A2: assume $-1-\omega$ and $3+\omega$ are yellow and $2-\omega$ and ω are orange.
The half-turn $z \mapsto 2-z$ about 1 maps the row with its colors, the points $-\omega, 1-\omega, 1+\omega, 2+\omega$, and the corner points of this case to themselves, and it maps key triples to key triples. So whenever the assumptions made so far are invariant under it, a statement proved from them also holds for its image: apply the same argument to the coloring $z \mapsto \chi(2-z)$, where χ is the given coloring. It is admissible, since the half-turn is an isometry. (The image of an argument in $\Lambda \cup q\Lambda$ takes place in $\Lambda \cup (2+q\Lambda)$.)

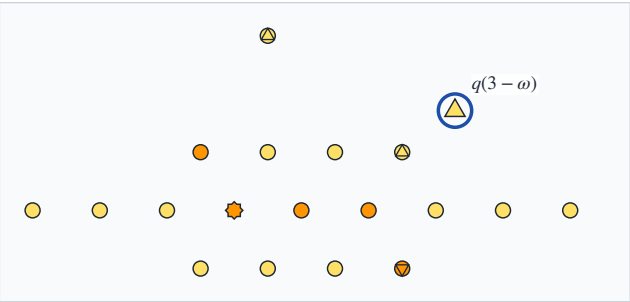




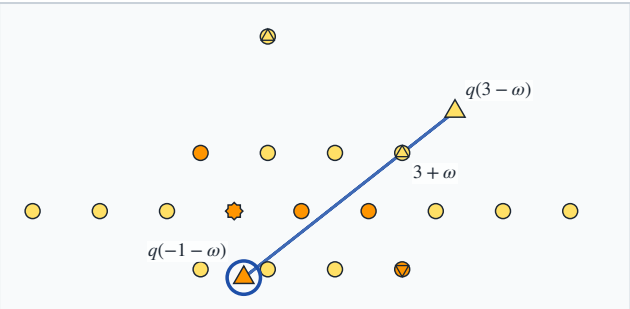
Step 1. $2+3\omega$ is yellow: $T(1,3,4)$ with orange $2-\omega$ and 2.
Check:

$(2+3\omega)-(2-\omega) = 4\omega$
$(2+3\omega)-2 = 3\omega$
$(2-\omega)-2 = -\omega$

Claim. $q(3-\omega)$ and $2-q(3-\omega)$ are orange.
Proof.

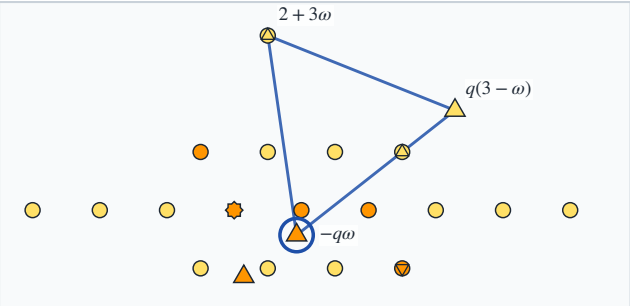


Suppose $q(3-\omega)$ is yellow.



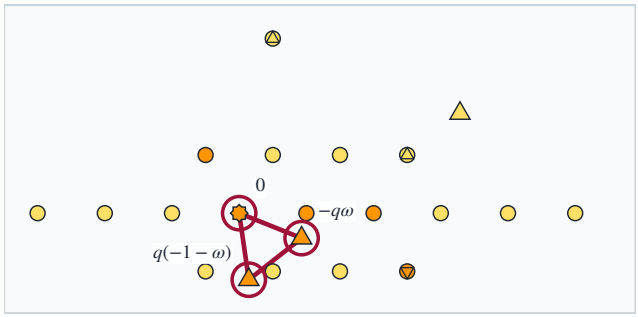
Step 1. $q(-1-\omega)$ is orange: $T(1,3,4)$ with yellow $3+\omega = q(2-\omega)$ and $q(3-\omega)$.
Check:

$q(2-\omega) = \dfrac{(8+5\omega)(2-\omega)}{7} = \dfrac{21+7\omega}{7} = 3+\omega$
$q(-1-\omega) - q(2-\omega) = -3q$
$q(-1-\omega) - q(3-\omega) = -4q$
$q(2-\omega) - q(3-\omega) = -q$

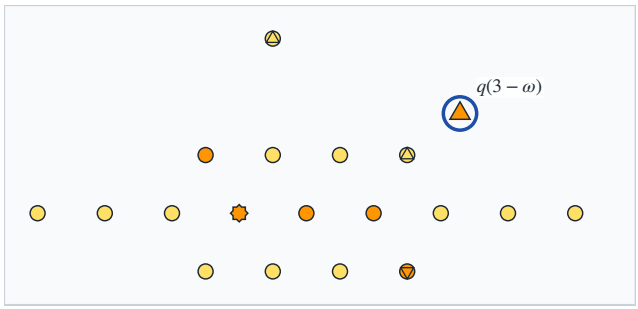


Step 2. $-q\omega$ is orange: $T(3)$ with yellow $2+3\omega = q(3+2\omega)$ and $q(3-\omega)$.
Check:

$q(3+2\omega) = \dfrac{(8+5\omega)(3+2\omega)}{7} = \dfrac{14+21\omega}{7} = 2+3\omega$
$(-q\omega) - q(3+2\omega) = -3q(1+\omega)$
$(-q\omega) - q(3-\omega) = -3q$
$q(3+2\omega) - q(3-\omega) = 3q\omega$



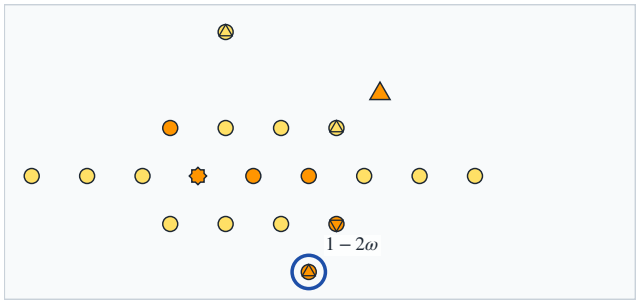
Contradiction: $q(-1-\omega)$, 0, $-q\omega$ are all orange and form $T(1)$.
Check:
$q(-1-\omega) - 0 = -q(1+\omega)$
$q(-1-\omega) - (-q\omega) = -q$
$0 - (-q\omega) = q\omega$



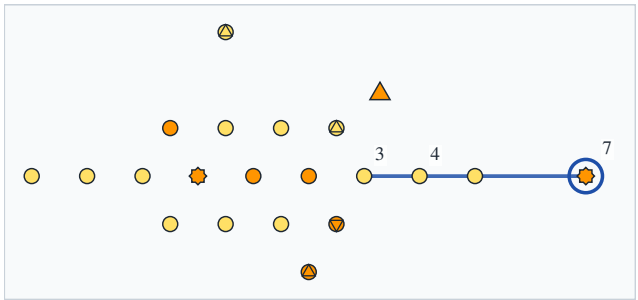
Claim: $q(3-\omega)$ and $2-q(3-\omega)$ are orange

So $q(3-\omega)$ is orange.
The same argument, applied to the coloring $z \mapsto \chi(2-z)$, shows that $2-q(3-\omega)$ is orange. □
The assumptions are now invariant under the half-turn.

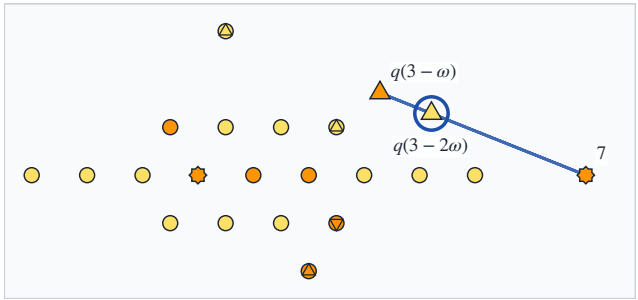
Claim. $1-2\omega$ and $1+2\omega$ are yellow.
Proof.



Suppose $1-2\omega$ is orange.



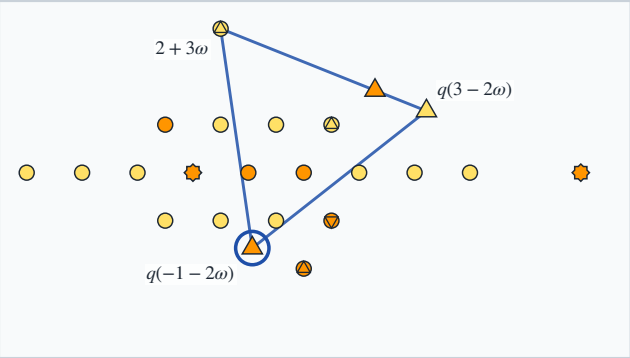
Step 1. 7 is orange: $T(1,3,4)$ with yellow 3 and 4.



Step 2. $q(3-2\omega)$ is yellow: $T(1,3,4)$ with orange $q(3-\omega)$ and $7 = q(3-5\omega)$.
Check:

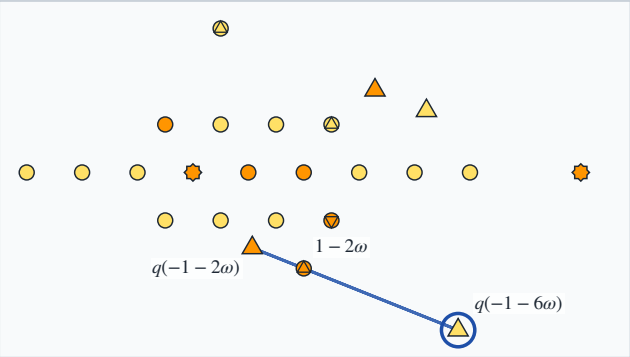
$$q(3-5\omega) = \frac{(8+5\omega)(3-5\omega)}{7} = \frac{49}{7} = 7$$

$q(3-2\omega) - q(3-\omega) = -q\omega$
$q(3-2\omega) - q(3-5\omega) = 3q\omega$
$q(3-\omega) - q(3-5\omega) = 4q\omega$



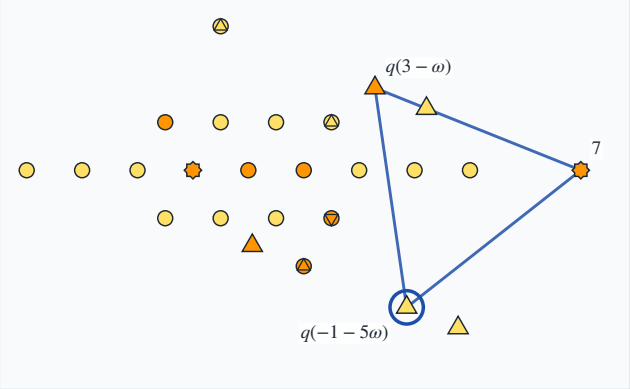
Step 3. $q(-1-2\omega)$ is orange: $T(4)$ with yellow $2+3\omega = q(3+2\omega)$ and $q(3-2\omega)$.
Check:

$$q(3+2\omega) = \frac{(8+5\omega)(3+2\omega)}{7} = \frac{14+21\omega}{7} = 2+3\omega$$
$$q(-1-2\omega) - q(3+2\omega) = -4q(1+\omega)$$
$$q(-1-2\omega) - q(3-2\omega) = -4q$$
$$q(3+2\omega) - q(3-2\omega) = 4q\omega$$



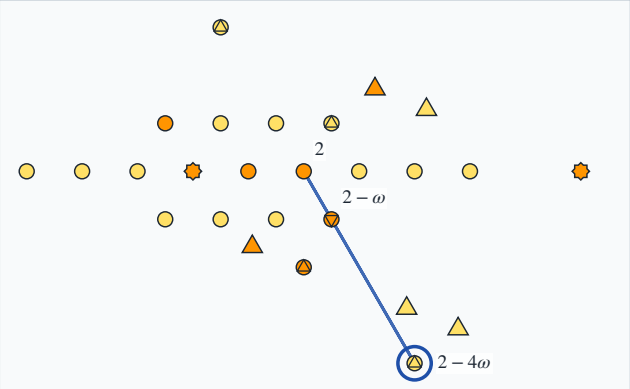
Step 4. $q(-1-6\omega)$ is yellow: $T(1,3,4)$ with orange $q(-1-2\omega)$ and $1-2\omega = q(-1-3\omega)$.
Check:

$$q(-1-3\omega) = \frac{(8+5\omega)(-1-3\omega)}{7} = \frac{7-14\omega}{7} = 1-2\omega$$
$$q(-1-6\omega) - q(-1-2\omega) = -4q\omega$$
$$q(-1-6\omega) - q(-1-3\omega) = -3q\omega$$
$$q(-1-2\omega) - q(-1-3\omega) = q\omega$$



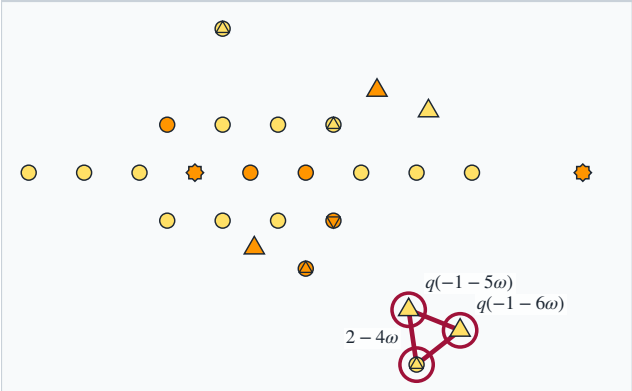
Step 5. $q(-1-5\omega)$ is yellow: $T(4)$ with orange $q(3-\omega)$ and $7 = q(3-5\omega)$.
Check:

$$q(3-5\omega) = \frac{(8+5\omega)(3-5\omega)}{7} = \frac{49}{7} = 7$$
$$q(-1-5\omega) - q(3-\omega) = -4q(1+\omega)$$
$$q(-1-5\omega) - q(3-5\omega) = -4q$$
$$q(3-\omega) - q(3-5\omega) = 4q\omega$$



Step 6. $2-4\omega$ is yellow: $T(1,3,4)$ with orange $2-\omega$ and 2.
Check:

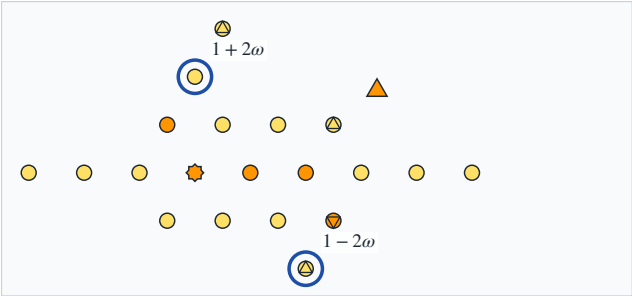
$$(2-4\omega) - (2-\omega) = -3\omega$$
$$(2-4\omega) - 2 = -4\omega$$
$$(2-\omega) - 2 = -\omega$$



Contradiction: $2-4\omega = q(-2-6\omega)$, $q(-1-5\omega)$, $q(-1-6\omega)$ are all yellow and form $T(1)$.

Check:

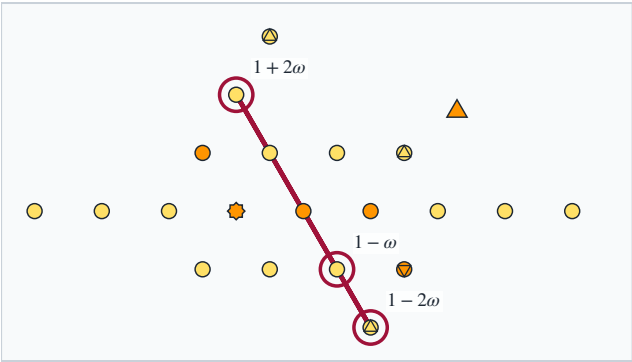
$$q(-2-6\omega) = \frac{(8+5\omega)(-2-6\omega)}{7} = \frac{14-28\omega}{7} = 2-4\omega$$
$$q(-2-6\omega) - q(-1-5\omega) = -q(1+\omega)$$
$$q(-2-6\omega) - q(-1-6\omega) = -q$$
$$q(-1-5\omega) - q(-1-6\omega) = q\omega$$



Claim: $1-2\omega$ and $1+2\omega$ are yellow

So $1-2\omega$ is yellow.
The same argument, applied to the coloring $z \mapsto \chi(2-z)$, shows that $1+2\omega$ is yellow. □

End of Case A2.



Contradiction: $1-2\omega$, $1-\omega$, $1+2\omega$ are all yellow and form $T(1,3,4)$.

Check:

$$(1-2\omega) - (1-\omega) = -\omega$$
$$(1-2\omega) - (1+2\omega) = -4\omega$$
$$(1-\omega) - (1+2\omega) = -3\omega$$

Since both cases lead to a contradiction, there is no maximal monochromatic run of exactly three points. □

3.2. **Rows.** Let L be a unit triangular lattice and u a long direction of L (Lemma 2.6), and put $v = -u\omega^2 = u(1+\omega)$. The three directions of L are u, v, and $v-u = u\omega$. The *rows* of L are its lines in direction u, and the row after R is $R+v$.

Lemma 3.4 (four apart)**.** *For every $z \in L$ we have $\chi(z+4u) \neq \chi(z)$ and $\chi(z+8u) = \chi(z)$.*

Proof. By Lemmas 2.4 and 2.5 every run in direction u has three or four points, and by Theorem 3.1 not three. So each row is colored with four orange points, four yellow points, and so on, repeating. □

Lemma 3.5 (consecutive rows)**.** *Let R be a row of L. Then either $\chi(z+v) \neq \chi(z)$ for every $z \in R$, or $\chi(z+v-u) \neq \chi(z)$ for every $z \in R$. We say that the pair of rows $(R, R+v)$ has* type *v or $v-u$ accordingly.*

Proof. Let $t, t+u, t+2u, t+3u$ be an orange run in R. Each of $t+v$, $t+u+v$, $t+2u+v$ forms a copy of $T(1)$ with two consecutive points of the run, so all three are yellow. By Lemma 3.4 the yellow run of $R+v$ containing them has exactly four points: it is either $t+v, \dots, t+3u+v$ or $t-u+v, \dots, t+2u+v$. In the first case $\chi(z+v) \neq \chi(z)$ for $z = t, \dots, t+3u$, in the second $\chi(z+v-u) \neq \chi(z)$. Since Lemma 3.4 flips the colors of both rows under $z \mapsto z+4u$, the same relation holds for every $z \in R$. □

3.3. **All pairs of consecutive rows have the same type.** Fix a row R. If $z, z+u \in R$ have the same color then the point $z+v$ above them and the point $z+u-v$ below them both have the other color. The remaining points of $R \pm v$, above or below a pair $z, z+u$ with $\chi(z) \neq \chi(z+u)$, are *open*. They come in vertically aligned *open pairs* $\{z+v,\ z+u-v\}$ (Figure 15). For example, in Figure 15 the row is $R = \mathbb{Z}+\omega$, $u = 1$, and $v = 1+\omega$. The orange points $z = -2+\omega$ and $z+u = -1+\omega$ force $z+v = -1+2\omega$ and $z+u-v = -2$ to be yellow (blue triangles). The points ω and $1+\omega$ have different colors, so $1+2\omega$ and 0 form an open pair (dashed).

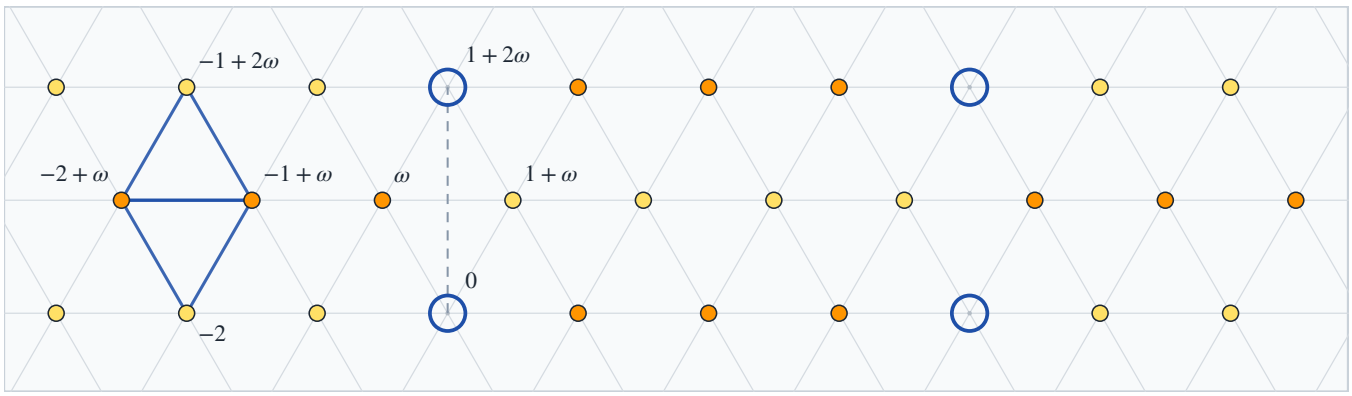


FIGURE 15. Open pairs at the row $\mathbb{Z}+\omega$

Lemma 3.6. *An open pair $\{z+v,\ z+u-v\}$ at the row R is monochromatic exactly when the pairs of rows $(R-v, R)$ and $(R, R+v)$ have different types.*

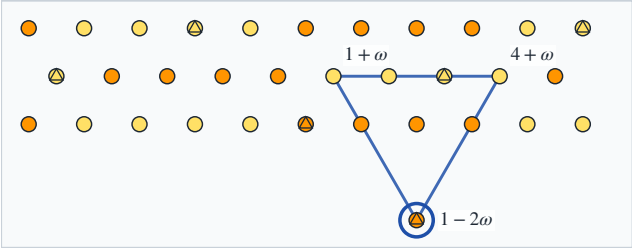
Proof. Let $c = \chi(z)$, so $\chi(z+u) \neq c$. If $(R, R+v)$ has type v, then $\chi(z+v) \neq c$. If it has type $v-u$, then $\chi(z+v) \neq \chi(z+u)$, so $\chi(z+v) = c$. Let $y = z+u-v \in R-v$. If $(R-v, R)$ has type v, then $\chi(y) \neq \chi(y+v) = \chi(z+u)$, so $\chi(y) = c$. If it has type $v-u$, then $\chi(y) \neq \chi(y+v-u) = \chi(z)$, so $\chi(y) \neq c$. Hence $\chi(z+v) = \chi(y)$ exactly when the two types differ. □

Theorem 3.7. *In an admissible coloring of the plane, all pairs of consecutive rows of a unit triangular lattice have the same type. Equivalently, every open pair has two colors.*

Proof. The two statements are equivalent by Lemma 3.6. Suppose that some open pair is monochromatic. Translating along the row, exchanging the colors, and reflecting in a line perpendicular to the rows if necessary, we may assume that $L = \Lambda$, $u = 1$, that the row $R = \mathbb{Z}+\omega$ is orange exactly at the points $a+\omega$ with $a \equiv 5, 6, 7, 0$, and that the open pair $\{1+2\omega,\ 0\}$ is orange. By copies of $T(1)$ and Lemma 3.4 the neighboring rows are then determined: $\mathbb{Z}$ is orange exactly at $a \equiv 0, 1, 2, 3$, and $\mathbb{Z}+2\omega$ is orange exactly at $a+2\omega$ with $a \equiv 1, 2, 3, 4$ (all congruences are modulo 8, see Figure 16). We derive a contradiction from these three rows using only Λ and $q\Lambda$, with the conventions of Section 3.1. One case split, on the point $q(-1+\omega)$, suffices, and the deduction used in both cases is listed first. As $\bar{q}\Lambda$ is not used, the point 0, which also lies in $\bar{q}\Lambda$, is drawn as a point of Λ and $q\Lambda$ (a circle with an inscribed upward triangle), and other points of $\Lambda \cap \bar{q}\Lambda$ as plain circles.



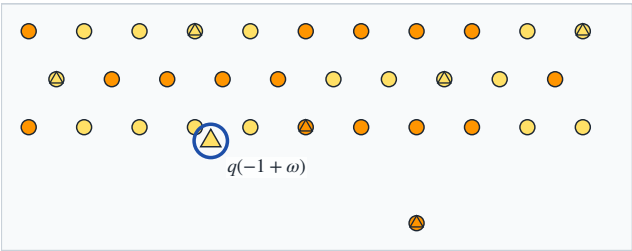
FIGURE 16. A monochromatic open pair

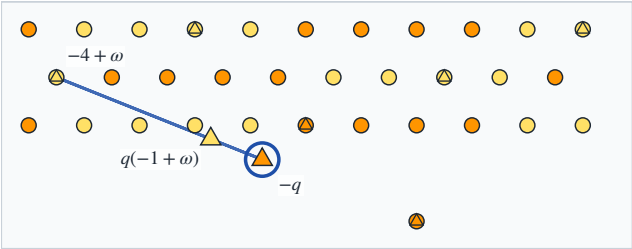


Step 1. $1-2\omega$ is orange: $T(3)$ with yellow $1+\omega$ and $4+\omega$.
Check:

$(1-2\omega)-(1+\omega)=-3\omega$
$(1-2\omega)-(4+\omega)=-3(1+\omega)$
$(1+\omega)-(4+\omega)=-3$

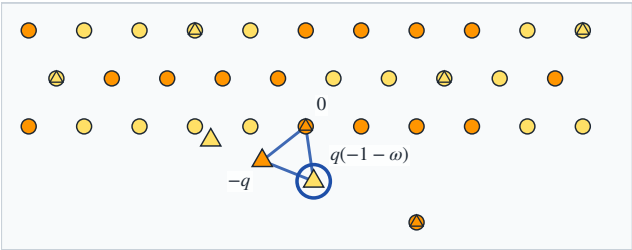
Case B1: assume $q(-1+\omega)$ is yellow.





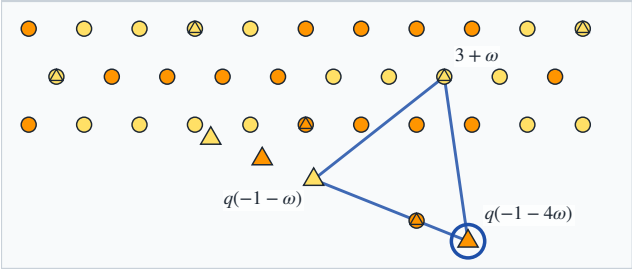
Step 1. $-q$ is orange: $T(1,3,4)$ with yellow $-4+\omega=q(-1+4\omega)$ and $q(-1+\omega)$.
Check:

$$q(-1+4\omega)=\frac{(8+5\omega)(-1+4\omega)}{7}=\frac{-28+7\omega}{7}=-4+\omega$$
$(-q)-q(-1+4\omega)=-4q\omega$
$(-q)-q(-1+\omega)=-q\omega$
$q(-1+4\omega)-q(-1+\omega)=3q\omega$



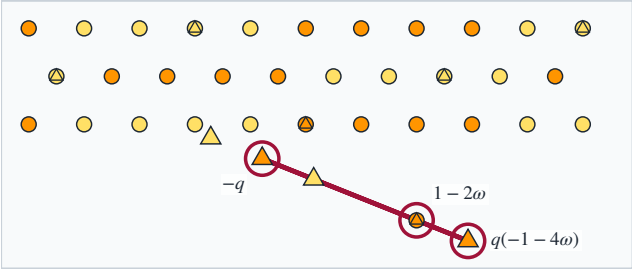
Step 2. $q(-1-\omega)$ is yellow: $T(1)$ with orange $-q$ and 0.
Check:

$q(-1-\omega)-(-q)=-q\omega$
$q(-1-\omega)-0=-q(1+\omega)$
$(-q)-0=-q$



Step 3. $q(-1-4\omega)$ is orange: $T(3)$ with yellow $q(-1-\omega)$ and $3+\omega = q(2-\omega)$.
Check:

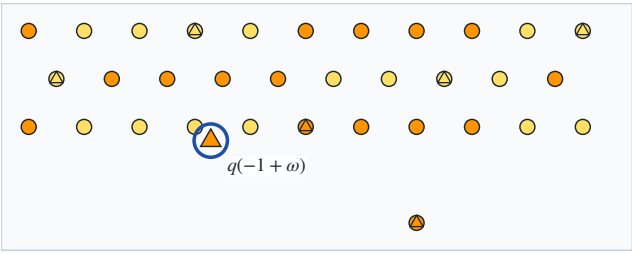
$$q(2-\omega) = \frac{(8+5\omega)(2-\omega)}{7} = \frac{21+7\omega}{7} = 3+\omega$$
$$q(-1-4\omega) - q(-1-\omega) = -3q\omega$$
$$q(-1-4\omega) - q(2-\omega) = -3q(1+\omega)$$
$$q(-1-\omega) - q(2-\omega) = -3q$$

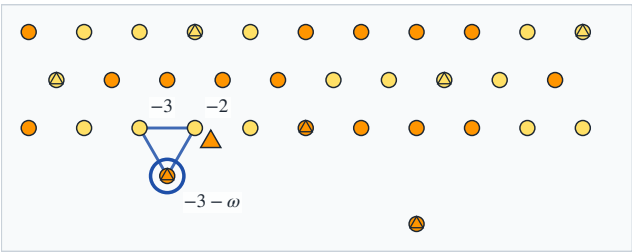


Contradiction: $-q$, $1-2\omega = q(-1-3\omega)$, $q(-1-4\omega)$ are all orange and form $T(1,3,4)$.
Check:

$$q(-1-3\omega) = \frac{(8+5\omega)(-1-3\omega)}{7} = \frac{7-14\omega}{7} = 1-2\omega$$
$$(-q) - q(-1-3\omega) = 3q\omega$$
$$(-q) - q(-1-4\omega) = 4q\omega$$
$$q(-1-3\omega) - q(-1-4\omega) = q\omega$$

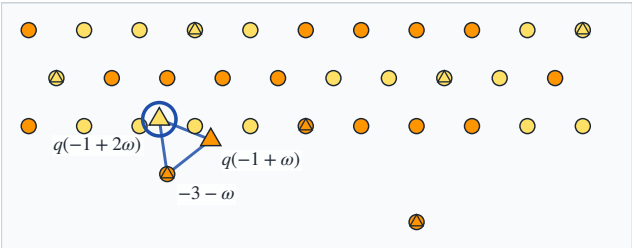
Case B2: assume $q(-1+\omega)$ is orange.





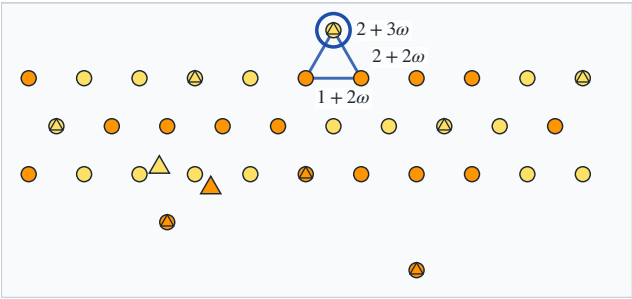
Step 1. $-3-\omega$ is orange: $T(1)$ with yellow -3 and -2.
Check:

$$(-3-\omega) - (-3) = -\omega$$
$$(-3-\omega) - (-2) = -(1+\omega)$$
$$(-3) - (-2) = -1$$



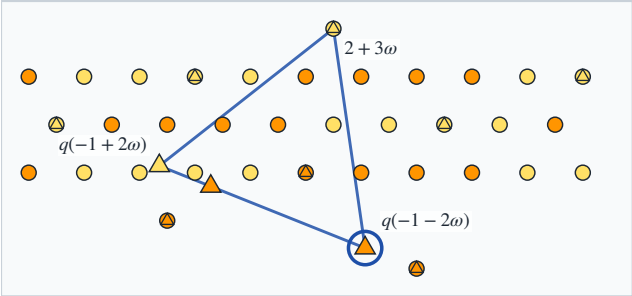
Step 2. $q(-1+2\omega)$ is yellow: $T(1)$ with orange $-3-\omega = q(-2+\omega)$ and $q(-1+\omega)$.
Check:

$$q(-2+\omega) = \frac{(8+5\omega)(-2+\omega)}{7} = \frac{-21-7\omega}{7} = -3-\omega$$
$$q(-1+2\omega) - q(-2+\omega) = q(1+\omega)$$
$$q(-1+2\omega) - q(-1+\omega) = q\omega$$
$$q(-2+\omega) - q(-1+\omega) = -q$$



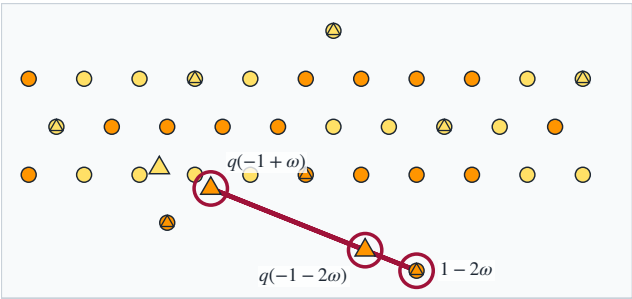
Step 3. $2+3\omega$ is yellow: $T(1)$ with orange $1+2\omega$ and $2+2\omega$.
Check:

$$(2+3\omega) - (1+2\omega) = 1+\omega$$
$$(2+3\omega) - (2+2\omega) = \omega$$
$$(1+2\omega) - (2+2\omega) = -1$$



Step 4. $q(-1-2\omega)$ is orange: $T(4)$ with yellow $q(-1+2\omega)$ and $2+3\omega = q(3+2\omega)$.
Check:

$$q(3+2\omega) = \frac{(8+5\omega)(3+2\omega)}{7} = \frac{14+21\omega}{7} = 2+3\omega$$
$$q(-1-2\omega) - q(-1+2\omega) = -4q\omega$$
$$q(-1-2\omega) - q(3+2\omega) = -4q(1+\omega)$$
$$q(-1+2\omega) - q(3+2\omega) = -4q$$



Contradiction: $q(-1+\omega)$, $q(-1-2\omega)$, $1-2\omega = q(-1-3\omega)$ are all orange and form $T(1,3,4)$.
Check:

$$q(-1-3\omega) = \frac{(8+5\omega)(-1-3\omega)}{7} = \frac{7-14\omega}{7} = 1-2\omega$$
$$q(-1+\omega) - q(-1-2\omega) = 3q\omega$$
$$q(-1+\omega) - q(-1-3\omega) = 4q\omega$$
$$q(-1-2\omega) - q(-1-3\omega) = q\omega$$

Neither case is possible, so no open pair is monochromatic. □

3.4. **Equal colors at distance eight.**

Proposition 3.8. *In an admissible coloring of the plane, any two points at distance* 8 *have the same color.*

Proof. Let $|p-p'| = 8$ and take a unit triangular lattice $L \ni p$ in which $p'-p = 8w$ for a direction w of L. By Theorem 3.7 all pairs of consecutive rows of L have the same type $t \in \{v, v-u\}$, so by Lemma 3.5 $\chi(z+t) \neq \chi(z)$ for every $z \in L$, and hence $\chi(z+8t) = \chi(z)$. By Lemma 3.4, $\chi(z+8u) = \chi(z)$. The third direction is $t-u$ or $t+u$, and $\chi(z+8t\pm 8u) = \chi(z)$ as well. So $\chi(p') = \chi(p)$ whichever direction w is. □

Theorem 1.3. *For every $a > 0$, every two-coloring of $\mathbb{R}^2$ contains $T(a, 3a, 4a)$.*

Proof. By scaling, we may assume $a = 1$. Suppose χ is admissible, and let A, B, C be collinear with $|A-B| = 1$, $|B-C| = 3$, and $|A-C| = 4$. On one side of the line, the circles of radius 8 about A and B meet in a point P, and those about B and C in a point Q (Figure 17). Proposition 3.8 gives $\chi(A) = \chi(P) = \chi(B) = \chi(Q) = \chi(C)$, so $\{A, B, C\}$ is a monochromatic $T(1,3,4)$, a contradiction. □

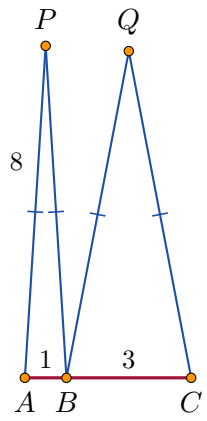


FIGURE 17. The end of the proof of Theorem 1.3

4. Use of AI

The exploration purely on the triangular grid, including the neighborhood argument, was done while writing my thesis [Yan25]. During that work, the idea of overlapping triangular lattices came to my mind as a potential direction, which the thesis mentions in its conclusion: I was driven by the intuition that the structure theorem is far too limiting, that several lattices have to be combined, and that the constraints would then accumulate quickly enough to force a contradiction. GPT-6 Astra discovered that after the overlap the coloring is 8-periodic, and found Theorems 3.1 and 3.7 as a way to split the proof of this periodicity. It also calculated the coordinates of the points needed and derived the proofs. Claude Opus 5.5 helped to find a shorter argument, in particular the use of the mirrored lattice $\bar{q}\Lambda$ in the symmetry arguments.

Acknowledgements

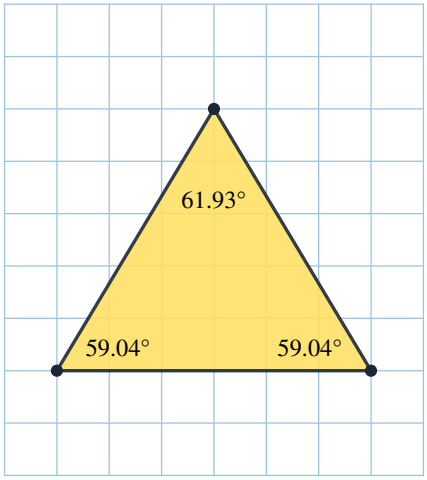
I am grateful to my thesis supervisor Jan Kynčl, whose suggestion to look at this problem created the spark for this work, and for his guidance during my thesis work, which made this paper possible.

Dedication

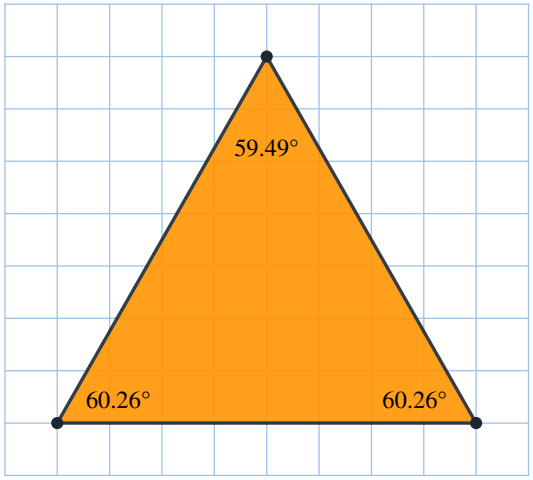
This work is dedicated to the memory of the extraordinary Kharkiv teacher, computer scientist, mathematician, and wartime volunteer Yulia Zdanovska[1] (*04.05.2000–†03.03.2022), who opened my eyes to the wonder of combinatorics, graphs, and algorithms. She spent countless hours patiently teaching me to solve problems in these fields when we were preparing for mathematics and informatics olympiads together, back in the carefree days of our childhood.

I remember us reading a paper on the chromatic number of space while sitting on the Giant Bench in Holosiiv Park, two and a half meters above the ground. It was a sunny, calm day in the forest by the lake full of ducks, and we were eating improvised sandwiches. That is when we discovered that we drew equilateral triangles on grid paper in different ways: I used isosceles triangles with base 6 and height 5, while she used ones with base 8 and height 7. We made a bet on a chocolate bar over whose approximation was better, and calculated the angles. Mine came out as $(59.04°,\ 59.04°,\ 61.93°)$, hers as $(60.26°,\ 60.26°,\ 59.49°)$. She won the chocolate bar.

I wish there were more such afternoons, and more bets for her to win.



base 6, height 5



base 8, height 7

I am eternally grateful for her inspiration and feel honored to have known her so closely.

[1]For more about her life and legacy, see `https://teachforukraine.org/en/t4u-news/ne-hochu-v-garvard-hochu-buty-ministerkoyu-osvity-istoriya-zagybloyi-matematykyni-yuliyi-zdanovskoyi/`.

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[Yan25] Alina Yan. *Monochromatic configurations in two-colored plane*. Bachelor's thesis. Charles University, Prague, 2025. Available from: https://dspace.cuni.cz/handle/20.500.11956/202494.